Formal Mathematical Framework

For Quantifying Logical Coherence in Natural Language Arguments

Companion to *Purposeless Efficiency*

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Abstract

This document presents the fully formal mathematical framework for a quantitative idea tester: a system that measures the internal logical coherence of natural language arguments. We provide rigorous definitions from propositional and first-order logic, paraconsistent logic, abstract and structured argumentation theory, probabilistic coherence measures, embedding-space geometry, and information theory. This expanded edition serves as a complete introductory course on all mathematics required to understand the ontological analysis of value as persuasion. Algorithms, complexity results, and composition rules for a composite coherence score are stated with full mathematical precision, paired with extensive intuitive explanations, worked examples, and connections to the core theory.

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1 Introduction: Value as Persuasion

Before diving into formal mathematics, we establish the conceptual foundation: **value is persuasion**. This is the thesis that a business proposition's durability—its ability to survive critical scrutiny and generate sustained market demand—is fundamentally determined by the logical coherence of its justification.

Consider a startup that claims to have "solved the transportation problem." The value the market assigns to this claim depends not merely on empirical efficacy, but on whether the argument *hangs together*. Contradictions in the reasoning, gaps in the logic, or circular dependencies in the justification undermine confidence—they reveal that the proposition is not genuinely understood or thought through. Conversely, an argument whose premises support the conclusion, whose components reinforce each other, and which avoids contradiction possesses a kind of durability. It is more persuasive precisely because it is more coherent.

The coherence engine measures this through a six-layer pipeline:

1. **Argument parsing:** Extract claims, premises, and their relationships from natural language.
2. **Contradiction detection:** Identify logical conflicts using NLI and paraconsistent logic.
3. **Dung's semantics:** Model argument attack/defense relationships as an abstract argumentation framework.
4. **Embedding coherence:** Measure semantic clustering and geometric relationships in embedding space.
5. **Compression analysis:** Quantify how much the argument compresses when treated as a unified whole.
6. **Composite scoring:** Synthesize all signals into a single coherence measure.

Each layer measures coherence from a different angle. The framework is conservative: we do not declare an argument coherent unless it passes multiple independent tests. This document teaches you the mathematics that makes each test possible.

2 Propositional and First-Order Consistency

The bedrock of logical coherence is the absence of contradiction. To formalize this, we must build up from the simplest logical language—propositional logic—and then extend to statements about objects, properties, and relationships using first-order logic. These sections teach you how to represent arguments mathematically and detect when they contradict themselves.

2.1 Propositional Logic: The Foundation

2.1.1 Why Propositional Logic Matters

Propositional logic is the study of how simple statements combine using logical connectives. Every natural language argument ultimately rests on claims that are either true or false in a given world. Propositional logic captures the essential structure: how do the truth values of basic claims determine the truth values of compound claims?

Consider this argument:

- Claim A: "Our product is cheaper than competitors."
- Claim B: "Our product has superior quality."
- Claim C: "The market will prefer our product."

In natural language, we might reason: "If A and B, then C." But what does this reasoning structure really mean? Propositional logic provides the vocabulary to make this precise. We ask: under what combinations of truth values for A, B, and C is the claim "If A and B, then C" itself true? This is foundational for detecting when an argument is self-contradictory.

2.1.2 Definitions and Truth Tables

Definition 2.1 (Propositional Language)**.** *A propositional language is a set of atomic propositions* $\mathcal{P} = \{p_1, p_2, \ldots, p_n\}$*, along with logical connectives:*

- *Negation:* $\neg p$ *("not p")*
- *Conjunction:* $p \wedge q$ *("p and q")*
- *Disjunction:* $p \vee q$ *("p or q")*
- *Conditional:* $p \to q$ *("if p then q")*
- *Biconditional:* $p \leftrightarrow q$ *("p if and only if q")*

Formulas are built recursively: each atomic proposition is a formula; if ϕ *and* ψ *are formulas, so are* $\neg\phi$*,* $(\phi \wedge \psi)$*,* $(\phi \vee \psi)$*,* $(\phi \to \psi)$*, and* $(\phi \leftrightarrow \psi)$*.*

Definition 2.2 (Truth Assignment and Models)**.** *A truth assignment* $v : \mathcal{P} \to \{\top, \bot\}$ *maps each atomic proposition to either true* ($\top$) *or false* ($\bot$)*. This assignment is extended inductively to all formulas:*

$$v(\neg\phi) = \top \iff v(\phi) = \bot \tag{1}$$
$$v(\phi \wedge \psi) = \top \iff v(\phi) = \top \text{ and } v(\psi) = \top \tag{2}$$
$$v(\phi \vee \psi) = \top \iff v(\phi) = \top \text{ or } v(\psi) = \top \text{ (or both)} \tag{3}$$
$$v(\phi \to \psi) = \top \iff v(\phi) = \bot \text{ or } v(\psi) = \top \tag{4}$$
$$v(\phi \leftrightarrow \psi) = \top \iff v(\phi) = v(\psi) \tag{5}$$

A truth assignment v *is called a* model *of a formula* ϕ *if* $v(\phi) = \top$*.*

2.1.3 Understanding the Conditional

The conditional $p \to q$ deserves special attention because it often trips up beginners. The truth table is:

p	q	$p \to q$
$\top$	$\top$	$\top$
$\top$	$\bot$	$\bot$
$\bot$	$\top$	$\top$
$\bot$	$\bot$	$\top$

Why is $p \to q$ true when p is false? Intuitively, the conditional says: "If the antecedent p holds, then the consequent q must hold." If p doesn't hold, the conditional makes no claim about q, so we deem it vacuously true. This is called the material conditional, and it differs from causal reasoning in natural language (where "if A then B" often carries an implicit causal connection). For our purposes, the material conditional is the correct formalization because we are only tracking logical dependence, not causality.

2.1.4 Example: Analyzing a Business Argument

Suppose a founder presents this argument for her startup:

1. p: "We have acquired 1,000 users in the first month."
2. q: "User retention is above 60%."
3. r: "We will achieve profitability within 18 months."
4. Claim: "If (p and q), then r." That is, $(p \wedge q) \to r$.

Now, an investor challenges her: "But you also said that the market is saturated (call this s), and if the market is saturated, we can't achieve profitability ($s \to \neg r$). These two claims are contradictory if p, q, and s are all true."

Let's check: suppose $v(p) = v(q) = v(s) = \top$. Then:

$$v((p \wedge q) \to r) = v(\top \to r) \tag{6}$$

$$v(s \to \neg r) = v(\top \to \neg r) = \text{depends on } v(r) \tag{7}$$

If the founder claims both $(p \wedge q) \to r$ and $s \to \neg r$, then the set $\{(p \wedge q) \to r, s \to \neg r, p, q, s\}$ is unsatisfiable (no truth assignment can make all of them true simultaneously). This is a contradiction that the coherence system detects.

2.1.5 Satisfiability and Unsatisfiability

Definition 2.3 (Satisfiability)**.** *A set of formulas $\Phi = \{\phi_1, \ldots, \phi_n\}$ is* satisfiable *if there exists a truth assignment v such that $v(\phi_i) = \top$ for all i. Otherwise, Φ is* unsatisfiable*. A formula ϕ is satisfiable if the set $\{\phi\}$ is satisfiable.*

Definition 2.4 (Logical Consequence)**.** *A formula ϕ is a* logical consequence *of a set of formulas Φ (written $\Phi \models \phi$) if every model of Φ is also a model of ϕ. In other words, whenever all formulas in Φ are true, ϕ must also be true.*

For our coherence system, we check whether the set of claims extracted from an argument is satisfiable. If it is unsatisfiable, the argument is self-contradictory at the propositional level.

2.1.6 Worked Example: Truth Table Construction

Let's construct the full truth table for the formula $(p \to q) \wedge (q \to r) \to (p \to r)$. This formula asserts the transitivity of implication.

p	q	r	$p \to q$	$q \to r$	$p \to r$	$(p \to q) \wedge (q \to r) \to (p \to r)$
$\top$	$\top$	$\top$	$\top$	$\top$	$\top$	$\top$
$\top$	$\top$	$\bot$	$\top$	$\bot$	$\bot$	$\top$
$\top$	$\bot$	$\top$	$\bot$	$\top$	$\top$	$\top$
$\top$	$\bot$	$\bot$	$\bot$	$\top$	$\bot$	$\top$
$\bot$	$\top$	$\top$	$\top$	$\top$	$\top$	$\top$
$\bot$	$\top$	$\bot$	$\top$	$\bot$	$\top$	$\top$
$\bot$	$\bot$	$\top$	$\top$	$\top$	$\top$	$\top$
$\bot$	$\bot$	$\bot$	$\top$	$\top$	$\top$	$\top$

Notice that this formula is true under all 8 possible truth assignments. Such a formula is called a *tautology*. It expresses a law of logic: the transitivity of implication always holds.

2.1.7 Thought Experiment: Detecting Inconsistency

Imagine an argument with four claims:

- A: "If we lower prices, demand increases."
- B: "Demand has increased."
- C: "We lowered prices."
- D: "We did not lower prices."

Formalize: $\phi_A = (\text{lower} \to \text{demand_up})$, $\phi_B = \text{demand_up}$, $\phi_C = \text{lower}$, $\phi_D = \neg\text{lower}$.

The set $\{\phi_A, \phi_B, \phi_C, \phi_D\}$ is unsatisfiable because ϕ_C and ϕ_D are directly contradictory. The coherence engine would flag this as an internal inconsistency at the propositional level.

2.2 First-Order Logic: Reasoning about Objects and Properties

2.2.1 Moving Beyond Propositions

Propositional logic treats statements as atomic units with no internal structure. But real arguments are richer. Consider:

- "All customers want value for money."
- "John is a customer."
- "Therefore, John wants value for money."

This is the classic form of deductive reasoning (a syllogism), but it cannot be captured in propositional logic. We need a language that talks about objects (John, customers), properties (wanting value), and relationships (being a customer). This is what first-order logic provides.

2.2.2 Syntax and Semantics

Definition 2.5 (First-Order Language)**.** *A first-order language consists of:*

- Variables*:* $x, y, z, \ldots$ *(ranging over objects)*
- Constants*:* $a, b, c, \ldots$ *(specific objects like "John")*

- Predicates*: $P, Q, R, \ldots$ with specified arities. For example, Customer(x) is a unary predicate (a property), and Wants(x, y) is a binary predicate (a relation).*
- Logical connectives*: $\neg, \wedge, \vee, \rightarrow, \leftrightarrow$ (same as propositional logic)*
- Quantifiers*: $\forall$ ("for all") and $\exists$ ("there exists")*

A formula is an expression built recursively:

- *Atomic formulas: $P(t_1, \ldots, t_n)$ where t_i are terms (variables or constants).*
- *If ϕ and ψ are formulas, so are $\neg\phi$, $(\phi \wedge \psi)$, etc.*
- *If ϕ is a formula and x is a variable, then $\forall x\, \phi$ and $\exists x\, \phi$ are formulas.*

Definition 2.6 (Interpretation)**.** *An* interpretation $\mathcal{I}$ *consists of:*

- *A domain D (the set of all objects).*
- *An assignment of each constant to an element of D.*
- *An assignment of each n-ary predicate to a subset of D^n (the set of tuples that satisfy the predicate).*

Given an interpretation $\mathcal{I}$ and a variable assignment σ (which maps variables to elements of D), the truth value of a formula is defined inductively. For quantifiers:

$$\mathcal{I}, \sigma \models \forall x\, \phi \quad \iff \quad \mathcal{I}, \sigma[x \mapsto d] \models \phi \text{ for all } d \in D \tag{8}$$

$$\mathcal{I}, \sigma \models \exists x\, \phi \quad \iff \quad \mathcal{I}, \sigma[x \mapsto d] \models \phi \text{ for some } d \in D \tag{9}$$

where $\sigma[x \mapsto d]$ is the variable assignment that agrees with σ on all variables except x, which is mapped to d.

2.2.3 Example: The Syllogism Formalized

Recall the argument:

- "All customers want value for money."
- "John is a customer."
- "Therefore, John wants value for money."

Formalize:

$$\phi_1 : \forall x\, (\text{Customer}(x) \rightarrow \text{WantsValue}(x)) \tag{10}$$

$$\phi_2 : \text{Customer}(\text{John}) \tag{11}$$

$$\phi_3 : \text{WantsValue}(\text{John}) \tag{12}$$

Now consider an interpretation $\mathcal{I}$ where:

- Domain: $D = \{\text{John}, \text{Alice}, \text{Bob}, \ldots\}$
- Customer(x) is true for John, Alice, and all others.

- WantsValue(x) is true for John, Alice, and all others.

Under this interpretation and any variable assignment, we have $\mathcal{I} \models \phi_1$, $\mathcal{I} \models \phi_2$, and $\mathcal{I} \models \phi_3$. Moreover, in *any* interpretation where ϕ_1 and ϕ_2 are true, ϕ_3 must also be true. This is why the argument is *logically valid*: the conclusion follows necessarily from the premises.

For the coherence system, this means: if an argument makes logical claims (universal statements, existential claims) and we can show that they are all simultaneously satisfiable, the argument avoids contradiction at the first-order level.

2.2.4 Gödel's Incompleteness and Its Implications

One of the deepest results in logic is Gödel's Incompleteness Theorem. We state a simplified version:

Theorem 2.1 (Gödel Incompleteness (Intuitive Version))**.** *For any formal system strong enough to express arithmetic, there exist true statements (true in the standard model) that cannot be proven from the axioms of the system. Moreover, the consistency of the system cannot be proven from within the system itself.*

Why does this matter for coherence measurement? It tells us that *no formal system can be both complete and consistent.* If we insist on consistency (absence of contradiction), we must accept that some true statements lie outside the reach of our logical rules. This has a profound implication for natural language arguments:

An argument may be internally consistent (no contradictions) yet fail to prove all relevant claims. This is not necessarily a defect—it reflects a fundamental limit of logical systems. What matters for coherence is not completeness but consistency plus the ability to derive the essential conclusions.

The coherence engine thus focuses on detecting inconsistency (a sign of broken reasoning) rather than verifying completeness (which is impossible). We measure whether an argument "holds together" (is consistent), not whether it exhausts all logical consequences.

2.2.5 Thought Experiment: Necessity vs. Contingency

Imagine a founder claims:

- "All who pay our subscription fee receive support."
- "Alice pays our subscription fee."
- "We will be acquired by Google within 5 years."

The first two claims entail that Alice receives support. But the third claim is not entailed by the first two, nor does it contradict them. It is *contingent*—it may or may not be true. A coherent argument should:

1. Derive the logical consequences of its premises (Alice receives support).
2. Avoid contradictions between premises and claimed consequences.
3. Distinguish between necessary consequences and contingent possibilities.

First-order logic is the tool that formalizes these distinctions.

3 Paraconsistent Logic

3.0.1 The Problem of Contradiction in Natural Language

In classical logic, a contradiction is catastrophic. If a set of formulas is contradictory, then *anything* logically follows from it—a principle known as the explosion principle or ex falso quodlibet.

Theorem 3.1 (Explosion Principle)**.** *If a set of formulas Φ is inconsistent (contradictory), then for any formula ψ, we have $\Phi \models \psi$. In other words, everything is a logical consequence of a contradiction.*

This is logically correct but practically catastrophic for natural language. Consider:

- Premise A: "Our product is state-of-the-art."
- Premise B: "Our product has outdated components."
- These premises are contradictory.
- By explosion, we can now conclude: "The Earth is flat." "Unicorns exist." Or any other statement.

But in reality, this argument is merely *partially incoherent*—it has a local contradiction, but that doesn't mean everything the founder said is gibberish. This is why natural language reasoning needs *paraconsistent logic*, which allows contradictions without explosion.

3.1 Classical Explosion and Its Limitation

3.1.1 When Does Explosion Matter?

The explosion principle says: from $\{P, \neg P\}$, derive anything. Formally:

$$\{p, \neg p\} \models \text{(any formula)} \tag{13}$$

Why? In classical logic, to prove ϕ, we assume $\neg\phi$ and derive a contradiction, then conclude ϕ. If we already have a contradiction, this proof technique becomes trivial—*any* formula serves as the consequent of a proof by contradiction.

For argument coherence, we want to allow local contradictions (which may reflect genuine tensions in an argument) without declaring the entire argument worthless. This motivates paraconsistent logic.

3.1.2 Example: Tension in Business Models

A founder pitches: "We are a bootstrapped company (Claim A: we self-fund), and we are actively raising venture capital (Claim B: we seek external funding)." These claims seem contradictory—how can you be bootstrapped while raising capital?

But the contradiction is resolvable: perhaps the company is bootstrapped today but pursuing growth financing for next year, or perhaps "bootstrapped" is relative (using modest external funding while maintaining control). The coherence system should:

1. Detect the tension between A and B.
2. Not assume that the entire argument explodes.
3. Look for other claims that might reconcile the tension.

Paraconsistent logic provides the formal framework for this.

3.2 Priest's Logic of Paradox (LP)

3.2.1 The Idea Behind LP

Priest's LP is the simplest paraconsistent logic. Its key insight: *a proposition can be both true and false simultaneously.* This is not a rejection of the law of non-contradiction but a refinement: we distinguish between truth values and truth states.

In classical logic, each proposition has exactly one truth value: true or false. In LP, a proposition can have a truth state:

- True only (the classical "true")
- False only (the classical "false")
- Both true and false (a paradoxical state)
- Neither true nor false (a truth value gap)

Definition 3.1 (LP Models)**.** *In LP, an interpretation assigns each atomic proposition to a subset of* $\{\top, \bot\}$*:*

- $v(p) = \{\top\}$*: p is true only.*
- $v(p) = \{\bot\}$*: p is false only.*
- $v(p) = \{\top, \bot\}$*: p is both true and false.*
- $v(p) = \emptyset$*: p is neither true nor false (a truth gap).*

Truth values for compound formulas extend inductively:

$$\top \in v(\neg\phi) \iff \bot \in v(\phi) \tag{14}$$
$$\bot \in v(\neg\phi) \iff \top \in v(\phi) \tag{15}$$
$$\top \in v(\phi \wedge \psi) \iff \top \in v(\phi) \text{ and } \top \in v(\psi) \tag{16}$$
$$\bot \in v(\phi \wedge \psi) \iff \bot \in v(\phi) \text{ or } \bot \in v(\psi) \tag{17}$$

and similarly for other connectives.

3.2.2 Why LP Avoids Explosion

In LP, the principle of explosion fails. From $\{p, \neg p\}$, we cannot derive an arbitrary formula q because:

- $v(p) = \{\top, \bot\}$ (both true and false)
- $v(\neg p) = \{\top, \bot\}$ (both true and false)
- These are both satisfied, but we cannot derive $v(q) = \{\top\}$ without additional information.

The contradiction is localized: we know p is problematic, but we don't automatically invalidate the entire system of reasoning.

3.2.3 Example: Detecting Paradox in an Argument

Suppose a pitch contains:

- p: "This market opportunity is overlooked by competitors."
- $\neg p$: "This market opportunity is well-known to competitors."

In classical logic, if the speaker somehow asserts both, the argument explodes—everything follows. In LP:

- We assign $v(p) = \{\top, \bot\}$ (the claim is paradoxically both asserted and denied).
- The system registers the contradiction locally.
- We can still ask: are there other claims in the argument that help resolve this tension?

The coherence measure will penalize the argument for the contradiction but won't declare it completely incoherent.

3.3 Da Costa's Logics of Formal Inconsistency (LFI)

3.3.1 A More Sophisticated Approach

Da Costa's LFI is more powerful than LP: it allows contradictions but tracks which propositions are "well-behaved" (consistent) and which are problematic. This is crucial for natural language because some sentences are rock-solid ("2+2=4") while others are genuinely ambiguous or disputed.

Definition 3.2 (Consistency Operator)**.** *In LFI, each proposition p has an associated* consistency operator $\circ p$ *(read: "p is consistent"). Intuitively, $\circ p$ is true if p is well-behaved (not paradoxical).*

The logic includes axioms governing consistency:

1. $\circ p \to (p \to (p \vee q))$ *(if p is consistent, then modus ponens applies to it)*
2. $\circ p \to (\neg p \to ((\neg p) \vee q))$ *(consistency enables logical inference)*
3. *The law of non-contradiction holds only for consistent propositions:* $\circ p \to \neg(p \wedge \neg p)$

3.3.2 Example: Resolving Ambiguity

In a pitch, the founder says:

- p: "Our technology is revolutionary." (assigned consistency: true)
- q: "Our technology is incremental." (assigned consistency: false—possibly paradoxical)
- r: "We will dominate the market."

With LFI, we can reason:

- Since $\circ p$ is true, we can apply classical reasoning to p: it is either fully true or fully false, and the law of non-contradiction holds.
- Since $\circ q$ is false, q might be both true and false, and we are cautious about inferring from it.
- Claim r depends on how we resolve the paradoxes. If we treat the paradoxical claim as less reliable, our confidence in r decreases.

This nuance is valuable for real arguments, which often contain claims of varying certainty and clarity.

3.3.3 Thought Experiment: Reconciling Contradictions

Imagine an AI company's argument:

- A: "Our AI is more accurate than human experts."
- B: "Our AI sometimes makes mistakes that humans catch."
- These seem contradictory, but they might both be true if "more accurate" is interpreted in aggregate or on specific tasks.

With LFI:

1. We note the apparent contradiction.
2. We assess whether both A and B can be assigned consistency.
3. If they can (via a suitable interpretation), the argument is coherent despite the tension.
4. If they cannot, we mark A and/or B as inconsistent and adjust our confidence.

4 Abstract and Structured Argumentation

4.0.1 From Logic to Argument Graphs

Classical logic asks: is a set of formulas consistent or inconsistent? But arguments are more structured. An argument is not just a bag of propositions; it is a directed graph where:

- Nodes are claims or arguments.
- Edges represent attack or support relationships.
- Resolving which arguments to accept requires choosing a coherent subset.

This is the domain of argumentation theory, pioneered by Dung. We model arguments as structured graphs and compute which sets of arguments can be "defended" against attack.

4.1 Dung's Abstract Argumentation Framework

4.1.1 Core Concepts

Definition 4.1 (Abstract Argumentation Framework (AAF))**.** *An abstract argumentation framework is a directed graph $\mathcal{AF} = (A, R)$ where:*

- *A is a set of arguments.*
- *$R \subseteq A \times A$ is a set of attack relations. $(a, b) \in R$ means argument a attacks argument b.*

The semantics of AAF is defined through the notion of extensions: which sets of arguments form coherent, defensible positions?

Definition 4.2 (Acceptability and Extensions)**.** *Given an AAF $\mathcal{AF} = (A, R)$ and a set of arguments $S \subseteq A$:*

A set S defends *an argument a if for every argument b that attacks a, some argument in S attacks b. Formally: for all $b \in A$ with $(b, a) \in R$, there exists $c \in S$ with $(c, b) \in R$.*

An admissible set *S is a set of arguments such that:*

1. *S is conflict-free: no two arguments in S attack each other.*
2. *S defends all its members: for each $a \in S$, S defends a.*

Definition 4.3 (Semantical Extensions)**.** *Under different semantics, the notion of an ideal extension varies:*

Grounded extension: *the minimal set of arguments that defend themselves. Start with arguments that attack no one else; add arguments defended only by these; iterate until fixed point.*

Preferred extensions: *maximal admissible sets. An admissible set S is preferred if there is no admissible set S' with $S \subset S'$.*

Stable extensions: *sets of arguments S that are conflict-free and attack every argument outside S.*

Complete extensions: *sets S that are admissible and defend every argument they defend (fixpoint condition).*

4.1.2 Intuition: Why Dung's Semantics Matters

In a complex argument with multiple premises and counterarguments, Dung's framework answers: which subset of claims should an agent accept? The answer is an extension—a set of claims that:

- Avoid self-contradiction (conflict-free).
- Are defended against all attacks (defensible).

For the coherence engine, we compute which extensions exist and measure how robust they are (do multiple extensions exist, or is the grounded extension unique?). An argument whose grounded extension is small, or where many competing extensions exist, is less coherent.

4.1.3 Worked Example: Argument Graph Construction and Resolution

Consider a debate about a startup's sustainability claims:

- Argument A1: "Our carbon footprint is below the industry average (claim: LowCarbonFootprint)."
- Argument A2: "Our manufacturing uses single-use plastics (claim: UsesSingleUsePlastic)."
- Argument A3: "Single-use plastics significantly increase carbon footprint (attack: A2 attacks LowCarbonFootprint)."
- Argument A4: "We offset all plastic with carbon credits (claim: OffsetsPlastic, attack: A4 attacks A2)."
- Argument A5: "Carbon offsets are unreliable (attack: A5 attacks OffsetsPlastic)."

Construct the attack graph:

$$R = \{(A2, \text{LowCarbonFootprint}), \quad (A3, A2), \tag{18}$$
$$(A4, A2), \quad (A5, A4)\} \tag{19}$$

Wait, let me reorganize. Arguments attack arguments:

$$R = \{(A2, A1), \quad (A4, A2), \quad (A5, A4)\} \tag{20}$$

That is:

- A2 attacks A1 (if we use single-use plastics, we can't have low carbon footprint).
- A4 attacks A2 (but we offset the impact).
- A5 attacks A4 (offsets are unreliable).

Now compute the grounded extension:

1. Identify arguments that are attacked: A1, A2, A4.
2. Identify attackers of attacked arguments: A2 attacks A1, A4 attacks A2, A5 attacks A4.
3. A5 is unattacked, so include it: $S_0 = \{A5\}$.
4. A5 attacks A4, so A4 is defended against. A4 attacks A2, so we can include A4: $S_1 = \{A5, A4\}$.
5. A4 attacks A2, so A2 is defended against. A2 attacks A1, so we can include A2: $S_2 = \{A5, A4, A2\}$.
6. A2 attacks A1, so A1 is defended against. A1 attacks nothing. We can include A1: $S_3 = \{A5, A4, A2, A1\}$.
7. No more additions. Grounded extension: $\{A5, A4, A2, A1\}$.

Interpretation: Under the grounded extension, we accept all arguments. This means the skeptical picture emerges: even though A5 challenges A4, and A4 challenges A2, the chain of defenses holds. The startup's claim of low carbon footprint (A1) survives scrutiny.

4.1.4 Complexity and Computation

Computing the grounded extension takes polynomial time. However, determining whether a given argument is in *all* preferred extensions (skeptical acceptance) is co-NP-complete. This means:

- For small arguments (hundreds of arguments), the computation is feasible.
- For very large arguments, we may need approximation algorithms.
- The framework allows trade-offs between computational cost and precision.

4.2 ASPIC+ Structured Argumentation

4.2.1 Adding Structure to Arguments

Dung's abstract framework treats arguments as black boxes. ASPIC+ (Assumption-based Argumentation plus Preferences) opens the black box and specifies:

- What makes an argument valid (its internal structure).
- How to compare arguments (preference orderings).
- How to resolve conflicts systematically.

Definition 4.4 (ASPIC+ Arguments)**.** *An ASPIC+ argument is a tree structure where:*

- *Leaf nodes are premises (basic facts).*

- *Internal nodes are inferences (rules).*
- *The root is the conclusion.*

Formally, an argument A for conclusion ϕ is:

1. *ϕ itself (if it is a premise).*
2. *A rule $Conc \leftarrow Prem_1, \ldots, Prem_n$ with corresponding arguments for each $Prem_i$.*

4.2.2 Example: Building an ASPIC+ Argument

Suppose we want to argue: "Our product will succeed in the market."

Premises:

- P1: "The market is growing at 20% annually."
- P2: "Our product is cheaper than competitors."
- P3: "Cheap products in growing markets succeed."

Inference rule:

$$\text{ProductSucceeds} \leftarrow \text{GrowingMarket}, \tag{21}$$
$$\text{CheaperThanCompetitors}, \tag{22}$$
$$\text{CheapSucceeds} \tag{23}$$

The argument tree:

```
        ProductSucceeds (conclusion)
           /         |         \
          /          |          \
    GrowingMarket  Cheaper  CheapSucceeds
       (P1)         (P2)        (P3)
```

This structure makes it clear which premises support the conclusion and what inferences are at play. A counterargument might attack any premise or the inference rule itself.

4.2.3 Undercutting Attacks

ASPIC+ distinguishes between two types of attacks:

Rebuttal: attacking the conclusion of an argument. ("You're wrong; the product won't succeed because the market is saturated.")

Undercut: attacking the validity of an inference. ("Even if the market is growing and the product is cheap, that rule is outdated; cheap products no longer guarantee success.")

Undercutting attacks are more subtle but crucial. They question not what was said but the reasoning that connects premises to conclusion.

4.3 Bipolar Argumentation Frameworks

4.3.1 Support and Attack

Most argumentation frameworks consider only attack relations. But arguments can also support each other:

- A1: "Our team has 20 years of experience in healthcare."
- A2: "We are therefore qualified to build a healthcare SaaS."
- A1 supports A2.

Bipolar frameworks (Cayrol and Lagasquie-Schiex) model both attack and support, capturing richer argumentative structures.

Definition 4.5 (Bipolar Argumentation Framework)**.** *A bipolar argumentation framework is a directed graph* $\mathcal{BAF} = (A, R^+, R^-)$ *where:*

- A *is a set of arguments.*
- $R^+ \subseteq A \times A$ *is the support relation.* $(a, b) \in R^+$ *means* a *supports* b*.*
- $R^- \subseteq A \times A$ *is the attack relation.* $(a, b) \in R^-$ *means* a *attacks* b*.*

Semantics for BAF is more complex because a supported argument gains strength from its supporters, and an attacked argument is weakened. The coherence implications are:

- An argument with many supporters and few attackers is coherent.
- An argument with few supporters and many attackers is incoherent.
- Circular support (A supports B supports A) is a form of coherence (mutual reinforcement) but must be distinct from begging the question.

4.3.2 Thought Experiment: Analyzing a Pitch's Network Structure

A founder's pitch has:

- Claim 1: "We have secured partnerships with major retailers."
- Claim 2: "Our unit economics are positive."
- Claim 3: "We will be profitable within 2 years."
- Claim 4: "Partnerships amplify distribution and reduce customer acquisition cost."
- Claim 5: "A skeptic says: Partnerships are illusions; they often fall apart."

Attack-support graph:

- Support: $1 \to 3$ (partnerships support profitability via claim 4).
- Support: $4 \to 3$ (the economic mechanism supports profitability).
- Support: $2 \to 3$ (positive unit economics support profitability).

- Attack: $5 \to 1$ (skepticism attacks the partnership claim).

In a bipolar framework, we see that:

- Claim 3 is supported by multiple sources (1, 2, 4), making it robust.
- Claim 1 is attacked by 5, so the pitch must defend it (by showing partnerships are binding, for example).
- The overall coherence depends on whether the supports outweigh the attacks.

5 Natural Language Inference Formalization

5.0.1 From Arguments to Semantic Relations

The frameworks above (propositional logic, argumentation theory) assume arguments are already parsed into premises and conclusions. But real arguments are in natural language: paragraphs of text. We need a bridge: how do we extract logical relations from sentences?

Natural Language Inference (NLI) is the task of determining the logical relationship between two sentences. Given a premise sentence and a hypothesis sentence, NLI asks: does the premise entail, contradict, or leave neutral the hypothesis?

5.0.2 Definition and Motivations

Definition 5.1 (Natural Language Inference Relations)**.** *Given two sentences s_1 (premise) and s_2 (hypothesis), the NLI relation is:*

Entailment: *$s_1 \Rightarrow s_2$ if the truth of s_1 guarantees the truth of s_2. Example: s_1 = "All birds have wings," s_2 = "Some birds have wings." We have entailment.*

Contradiction: *$s_1 \Rightarrow \neg s_2$ if the truth of s_1 guarantees the falsity of s_2. Example: s_1 = "The startup raised $10M," s_2 = "The startup raised no outside funding." We have contradiction.*

Neutral: *Neither entailment nor contradiction. Example: s_1 = "The startup is based in San Francisco," s_2 = "The startup makes hardware." The location says nothing about the product category.*

5.0.3 Worked Example: Analyzing an Argument

Consider a pitch:

- S_1: "We have acquired 10,000 users in three months."
- S_2: "We are the fastest-growing app in our category."
- S_3: "The market for productivity apps is growing at 15% annually."

Now assess NLI relations:

- $S_1 \Rightarrow S_2$? Acquiring 10,000 users in three months does *not necessarily* make you the fastest-growing. Competitors might have grown faster. Verdict: neutral (not entailment).
- $S_1 \Rightarrow \neg S_3$? Rapid user growth doesn't contradict market growth. Verdict: neutral.
- $S_2 \wedge S_3 \Rightarrow$ "We will be successful"? Being the fastest-growing in a growing market suggests but doesn't guarantee success. Verdict: not an entailment, but a strong support.

For the coherence engine:

- We extract pairwise NLI relations from all sentences.
- Contradictions lower coherence.
- Entailments that support claimed conclusions raise coherence.
- Neutral pairs contribute neutral information (neither boosting nor damaging coherence).

5.0.4 Computing NLI with Neural Networks

Modern NLI is computed using transformer-based neural networks (e.g., BERT, RoBERTa) trained on large datasets. These models learn to classify sentence pairs into entailment, neutral, or contradiction, though they are not logically transparent. The output is a confidence score:

$$\Pr(\text{entailment} \mid s_1, s_2) \in [0, 1]$$

For our coherence framework, we interpret these scores as estimates of the degree of entailment, using them as inputs to probabilistic measures.

5.0.5 Limitations and Caveats

NLI models are powerful but imperfect:

- They can miss subtle contradictions that require world knowledge.
- They sometimes report entailment when careful logical analysis would differ.
- They don't capture vagueness (a key feature of natural language).

The coherence system thus treats NLI as one of six signals, not the sole judge. If NLI and other measures agree on a contradiction, confidence is high; if they disagree, we flag the ambiguity.

6 Embedding Space Geometry

6.0.1 What Embeddings Capture

Sentence embeddings (vector representations of sentences in high-dimensional space) capture semantic meaning through geometry: sentences with similar meanings are close together, distant sentences have different meanings.

Why is this useful for coherence? Because coherent arguments cluster around a few central themes, while incoherent arguments scatter their claims across disparate semantic regions.

6.1 Sentence Embeddings

6.1.1 Dense Vector Representations

Definition 6.1 (Sentence Embedding)**.** *A sentence embedding is a function* emb : *Sentence* $\to \mathbb{R}^d$ *that maps natural language sentences to dense vectors in d-dimensional space (typically* $d \in [384, 1536]$*).*

Common embedders include:

- *BERT (bidirectional): uses masked language modeling to learn contextual representations.*
- *Sentence-BERT (SBERT): a variant fine-tuned for sentence-level similarity.*
- *OpenAI's Ada or Text-Embedding-3: models trained on large text corpora and aligned with human judgment.*

The choice of embedder affects the results. A domain-specific embedder (trained on startup pitches) might outperform a general-purpose model. But general embedders capture broad semantic relationships.

6.1.2 Intuition: Geometry of Meaning

Imagine a three-dimensional embedding space (easier to visualize than 512-dimensional). Claims from a coherent pitch cluster in a small region:

- "We have strong demand signals." $\rightarrow$ (0.8, 0.1, 0.5)
- "Customer acquisition cost is dropping." $\rightarrow$ (0.75, 0.15, 0.52)
- "Revenue is growing month-over-month." $\rightarrow$ (0.78, 0.08, 0.48)

All these vectors are close together (they all discuss market traction). But a claim like:

- "The board members have impressive CVs." $\rightarrow$ (0.1, 0.9, 0.2)

is far away (it discusses team credentials, not market traction). A truly coherent pitch would either cluster all claims or have clear sub-clusters with bridges connecting them.

6.2 Semantic Coherence via Embeddings

6.2.1 Computing Semantic Similarity

Definition 6.2 (Cosine Similarity)**.** *Given two vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^d$, the cosine similarity is:*

$$\cos(\mathbf{u}, \mathbf{v}) = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|} \in [-1, 1]$$

where $\mathbf{u} \cdot \mathbf{v}$ is the dot product and $\|\mathbf{u}\|$ is the Euclidean norm. When vectors are normalized (unit vectors), cosine similarity simplifies to the dot product.

Cosine similarity of 1 indicates identical direction (highly similar), 0 indicates orthogonal directions (independent), and -1 indicates opposite directions (contradictory).

6.2.2 Worked Example: Clustering Pitch Claims

Suppose we embed all claims in a pitch:

- C_1: "We have 50,000 active users." $\rightarrow \mathbf{e}_1$
- C_2: "Monthly active users grew 30% last month." $\rightarrow \mathbf{e}_2$
- C_3: "Our team includes veterans from Google and Apple." $\rightarrow \mathbf{e}_3$
- C_4: "We have closed enterprise deals with two Fortune 500 companies." $\rightarrow \mathbf{e}_4$

Compute pairwise similarities:

- $\cos(\mathbf{e}_1, \mathbf{e}_2) = 0.85$ (both discuss user growth; highly similar)
- $\cos(\mathbf{e}_1, \mathbf{e}_3) = 0.15$ (user numbers vs. team credentials; low similarity)
- $\cos(\mathbf{e}_1, \mathbf{e}_4) = 0.72$ (both discuss traction; moderately similar)
- $\cos(\mathbf{e}_3, \mathbf{e}_4) = 0.30$ (team vs. deals; low similarity)

The coherence system observes:

1. Claims C_1 and C_2 cluster together (user metrics).
2. Claim C_3 is an outlier (team background).
3. Claim C_4 moderately bridges user metrics and team strength.

A good pitch would either expand on why team credentials matter (bridge the gap between C_3 and others) or focus on one or two themes. The current arrangement suggests some lack of thematic unity.

6.3 Geometric Contradiction Signature

6.3.1 Detecting Semantic Opposition

When two claims have cosine similarity close to -1, they are semantically opposite. This is a geometric signature of contradiction.

Definition 6.3 (Geometric Contradiction)**.** *Two claims s_1 and s_2 exhibit a geometric contradiction if $\cos(\text{emb}(s_1), \text{emb}(s_2)) < -\tau$ for some threshold τ (typically 0.3 or 0.5).*

Example:

- s_1: "Our technology is cutting-edge and novel." $\rightarrow \mathbf{e}_1$
- s_2: "Our approach uses well-established, battle-tested methods." $\rightarrow \mathbf{e}_2$
- $\cos(\mathbf{e}_1, \mathbf{e}_2) \approx -0.6$ (opposite directions)

The geometric signature detects this tension. The coherence engine flags it and asks: are these genuinely contradictory, or can both be true (e.g., applying novel insights to established methods)?

6.4 Concept-Axis Reflection

6.4.1 Understanding Semantic Dimensions

In embedding spaces, certain directions correspond to semantic axes. For example:

- The "positive sentiment" axis points from negative words ("failure", "problem") toward positive words ("success", "opportunity").
- The "market size" axis points from small-market words toward large-market words.

Definition 6.4 (Concept-Axis Reflection)**.** *Given an embedding space and two sentences s_1 and s_2, a concept-axis reflection occurs if:*

1. $\text{emb}(s_1)$ *and* $\text{emb}(s_2)$ *project to opposite sides of a conceptual axis.*
2. *The projections are roughly equidistant from the axis.*

Mathematically, if axis direction is $\mathbf{a}$ *(unit vector) and projections are* $(\mathbf{e}_1 \cdot \mathbf{a})$ *and* $(\mathbf{e}_2 \cdot \mathbf{a})$*, then reflection is detected if* $(\mathbf{e}_1 \cdot \mathbf{a})$ *and* $(\mathbf{e}_2 \cdot \mathbf{a})$ *have opposite signs and similar magnitudes.*

6.4.2 Worked Example: Tension on the Sentiment Axis

Suppose a pitch contains:

- S_1: "The market is saturated with competitors." (projects negatively on sentiment axis)
- S_2: "We have a unique competitive advantage." (projects positively on sentiment axis)

The concept-axis reflection detects the tension: one claim emphasizes market difficulty, the other emphasizes advantage. Both could be true (hard market, but we're better), or they could reflect incoherent reasoning.

The coherence engine thus flags reflection and seeks bridges (additional claims that explain how advantage overcomes saturation, for example).

6.4.3 Thought Experiment: Domain-Specific Axes

In a healthcare pitch, a "regulatory burden" axis might be identified:

- Positive direction: claims emphasizing compliance, FDA approval, regulatory experience.
- Negative direction: claims emphasizing speed-to-market, disruption, circumventing regulation.

Reflections on this axis indicate tension: does the pitch want to emphasize regulatory rigor or regulatory agility? A coherent pitch should choose a consistent position (or explicitly reconcile the tension).

7 Probabilistic Coherence Measures

7.0.1 From Logical to Probabilistic

Logical frameworks (propositional logic, argumentation theory) are all-or-nothing: a set of formulas is either consistent or contradictory. But real arguments have uncertainty. A claim might be plausible but not certain; a contradiction might be mild rather than severe.

Probabilistic coherence measures assign numerical scores reflecting the degree to which a set of claims cohere. The foundational insight: coherence is the degree to which claims mutually support and explain each other.

7.1 Shogenji's Measure

7.1.1 The Idea

Shogenji's coherence measure is based on a simple intuition: a set of claims is coherent if they share a common cause or explanation.

Definition 7.1 (Shogenji Coherence)**.** *Given claims* $\{C_1, \ldots, C_n\}$ *with probabilities* $\Pr(C_i)$*, Shogenji defines coherence as:*

$$S(C_1, \ldots, C_n) = \frac{\Pr(C_1 \wedge \cdots \wedge C_n)}{\Pr(C_1) \cdots \Pr(C_n)}$$

This is the ratio of the joint probability to the product of marginal probabilities. If the claims are independent, this ratio is 1 (minimal coherence). If the claims are highly dependent (supporting each other), the ratio is large.

7.1.2 Numerical Example

Suppose a founder makes three claims:

- C_1: "The market is growing rapidly."
- C_2: "We have strong customer demand."
- C_3: "Our revenue is growing exponentially."

Assign baseline probabilities (estimated from historical data or market analysis):

- $\Pr(C_1) = 0.6$ (60% of markets grow rapidly)
- $\Pr(C_2) = 0.5$ (50% of startups have strong demand)
- $\Pr(C_3) = 0.4$ (40% of startups achieve exponential growth)

If these claims were independent:

$$\Pr(C_1 \wedge C_2 \wedge C_3) = 0.6 \times 0.5 \times 0.4 = 0.12$$

But suppose market data and industry knowledge suggest that when all three claims are true together, it's actually more likely than the independent assumption (because rapid market growth tends to drive both demand and revenue growth). Estimate:

$$\Pr(C_1 \wedge C_2 \wedge C_3) = 0.35$$

Then Shogenji coherence is:

$$S = \frac{0.35}{0.12} \approx 2.92$$

A score greater than 1 indicates positive coherence—the claims fit together better than independence would suggest.

7.1.3 Advantages and Limitations

Advantages:

- Simple, intuitive formula.
- Captures the idea that coherent claims are more likely together than separately.

Limitations:

- Requires accurate probability estimates, which are often unavailable.
- Sensitive to the choice of prior probabilities.
- Treats all claims symmetrically; doesn't distinguish between premises and conclusions.

7.2 Olsson's Measure

7.2.1 Comparative Approach

Olsson's measure is comparative: it asks not "how coherent are these claims?" but "are these claims more coherent than an alternative set?"

Definition 7.2 (Olsson Coherence)**.** *Given two sets of claims Φ and Ψ, Olsson defines:*

$$O(\Phi|\Psi) = \frac{\Pr(\Phi|E) - \Pr(\Psi|E)}{\Pr(\Phi|E) + \Pr(\Psi|E)}$$

where E is evidence. This measure ranges from -1 to 1: a score of 1 means Φ is vastly more coherent than Ψ given the evidence.

7.2.2 Practical Example

A pitch offers two competing narratives for growth:

Narrative A (Φ):

- We have strong product-market fit (high retention, NPS).
- We've achieved unit economic profitability.
- Therefore, we'll grow via word-of-mouth and organic channels.

Narrative B (Ψ):

- We have a partially tested product.
- We need to acquire users through paid marketing.
- Growth depends on maintaining CAC/LTV ratio below 3:1.

Given evidence E (market size, startup data, comparable companies), which narrative is more coherent?

Estimate posterior probabilities:

- $\Pr(\Phi|E) = 0.7$ (Narrative A is likely given observed evidence: high retention, profitable unit economics)
- $\Pr(\Psi|E) = 0.3$ (Narrative B is less likely; contradicts observed profitability)

Then:

$$O(\Phi|\Psi) = \frac{0.7 - 0.3}{0.7 + 0.3} = \frac{0.4}{1.0} = 0.4$$

This indicates moderate preference for Narrative A, suggesting the pitch is reasonably coherent in choosing this growth story.

7.3 Fitelson's Measure

7.3.1 Fixing Problematic Cases

Fitelson critiques Shogenji's measure for behaving counterintuitively in certain cases. He proposes:

Definition 7.3 (Fitelson Coherence)**.**

$$F(C_1, \ldots, C_n) = \frac{1}{n(n-1)} \sum_{i \neq j} \frac{\Pr(C_i|C_j) - \Pr(C_i)}{\Pr(C_i|C_j) + \Pr(C_i)}$$

This averages pairwise coherence over all claim pairs. For each pair, it measures whether one claim's probability increases given the other (positive coherence) or decreases (negative coherence).

7.3.2 Numerical Example

Using our three claims from Shogenji's example:

- C_1: Rapid market growth
- C_2: Strong customer demand
- C_3: Exponential revenue growth

Estimate conditional probabilities:

$$\Pr(C_1|C_2) = 0.75 \quad \text{(if demand is strong, market is likely growing)} \tag{24}$$
$$\Pr(C_2|C_1) = 0.70 \quad \text{(if market grows, demand may be strong)} \tag{25}$$
$$\Pr(C_1|C_3) = 0.80 \quad \text{(if revenue grows exponentially, market grows)} \tag{26}$$
$$\Pr(C_3|C_1) = 0.65 \quad \text{(market growth often leads to revenue growth)} \tag{27}$$
$$\Pr(C_2|C_3) = 0.85 \quad \text{(exponential revenue suggests strong demand)} \tag{28}$$
$$\Pr(C_3|C_2) = 0.60 \quad \text{(demand exists, but revenue growth is not guaranteed)} \tag{29}$$

With base rates $\Pr(C_1) = 0.6$, $\Pr(C_2) = 0.5$, $\Pr(C_3) = 0.4$:

$$\text{Pair } (C_1, C_2): \quad \frac{0.75 - 0.6}{0.75 + 0.6} = \frac{0.15}{1.35} \approx 0.111 \tag{30}$$
$$\text{Pair } (C_2, C_1): \quad \frac{0.70 - 0.5}{0.70 + 0.5} = \frac{0.20}{1.20} \approx 0.167 \tag{31}$$
$$\text{Pair } (C_1, C_3): \quad \frac{0.80 - 0.4}{0.80 + 0.4} = \frac{0.40}{1.20} \approx 0.333 \tag{32}$$
$$\text{Pair } (C_3, C_1): \quad \frac{0.65 - 0.6}{0.65 + 0.6} = \frac{0.05}{1.25} \approx 0.040 \tag{33}$$
$$\text{Pair } (C_2, C_3): \quad \frac{0.85 - 0.4}{0.85 + 0.4} = \frac{0.45}{1.25} \approx 0.360 \tag{34}$$
$$\text{Pair } (C_3, C_2): \quad \frac{0.60 - 0.4}{0.60 + 0.4} = \frac{0.20}{1.00} = 0.200 \tag{35}$$
$$\tag{36}$$

Average:

$$F = \frac{1}{6}(0.111 + 0.167 + 0.333 + 0.040 + 0.360 + 0.200) \approx 0.202$$

This positive value (ranging from -1 to 1) indicates that the claims cohere—each increases the probability of the others.

7.4 Roche's Measure (Recommended)

7.4.1 Theoretical Advantages

Roche's measure addresses shortcomings in both Shogenji and Fitelson by grounding coherence in explanatory power.

Definition 7.4 (Roche Coherence)**.**

$$R(C_1, \ldots, C_n|B) = \frac{n \Pr(C_1 \wedge \cdots \wedge C_n|B) - \sum_i \Pr(C_i|B)}{\sum_i \Pr(C_i|B)}$$

where B is background knowledge. This measure compares the joint probability of all claims to the sum of individual probabilities, normalized.

7.4.2 Why Roche Is Recommended

1. **Explanatory grounding:** Roche's measure connects coherence to how well a set of claims explains background facts.
2. **Principled normalization:** The normalization by sum of marginals avoids extreme values.
3. **Logical connection:** It recovers classical logical coherence (satisfiability) as a special case.

For the coherence engine, we use Roche's measure as the primary probabilistic coherence signal, supplemented by Shogenji for fast approximation.

7.4.3 Thought Experiment: Evaluating Coherence

A biotech pitch contains:

- C_1: "Our drug targets protein X, which is known to drive disease Y."
- C_2: "We've conducted Phase I trials showing safety."
- C_3: "The market for drugs targeting Y is \$5B annually."
- C_4: "We expect to reach market in 4 years."

Background knowledge B includes: biotech development timelines, disease epidemiology, regulatory pathways.

Assign probabilities given B:

- $\Pr(C_1|B) = 0.7$ (protein X is a known target)
- $\Pr(C_2|B) = 0.6$ (Phase I success rate)
- $\Pr(C_3|B) = 0.8$ (market size estimate is reasonable)
- $\Pr(C_4|B) = 0.5$ (4-year timelines are optimistic)
- $\Pr(C_1 \wedge C_2 \wedge C_3 \wedge C_4|B) = 0.25$ (all together: reasonable but not certain)

Compute Roche:

$$R = \frac{4 \times 0.25 - (0.7 + 0.6 + 0.8 + 0.5)}{0.7 + 0.6 + 0.8 + 0.5} = \frac{1.0 - 2.6}{2.6} \approx -0.62$$

A negative score suggests the claims are less coherent than we'd expect—perhaps the 4-year timeline clashes with the Phase I safety data (which, if it's already completed, suggests the company is further along than 4 years from market).

8 Information-Theoretic Measures

8.0.1 Compression and Meaning

A coherent argument is, in some sense, "compressible." If claims are independent, you need many words to describe them. If claims form a unified whole, supporting each other, you can explain them more concisely.

Information theory formalizes this. We measure coherence through:

- **Entropy:** the inherent unpredictability or diversity of claims.
- **Mutual information:** how much one claim tells you about another.
- **Compression ratio:** how much the description shrinks when claims are unified.

8.1 Shannon Entropy

8.1.1 Quantifying Uncertainty

Definition 8.1 (Shannon Entropy)**.** *The entropy of a probability distribution P over outcomes $X = \{x_1, \ldots, x_n\}$ is:*

$$H(X) = -\sum_i \Pr(x_i) \log_2 \Pr(x_i)$$

measured in bits. Higher entropy indicates greater uncertainty; lower entropy indicates more predictability.

8.1.2 Example: Entropy of Claims

Suppose a pitch claims:

- With probability 0.9: "We will reach profitability."
- With probability 0.1: "We will not reach profitability."

The entropy of this single binary claim is:

$$H = -0.9 \log_2(0.9) - 0.1 \log_2(0.1) \approx -0.9 \times (-0.152) - 0.1 \times (-3.322) \approx 0.137 + 0.332 = 0.469 \text{ bits}$$

This is low entropy (close to 0 bits), reflecting high confidence. The claim is predictable: profitability is very likely.

Now consider a rival pitch that claims:

- With probability 0.5: "We will reach profitability."
- With probability 0.5: "We will not reach profitability."

Entropy:

$$H = -0.5 \log_2(0.5) - 0.5 \log_2(0.5) = -0.5 \times (-1) - 0.5 \times (-1) = 1 \text{ bit}$$

This is maximum entropy for a binary outcome, reflecting maximum uncertainty. The pitch makes no confident claim about profitability—it is equally likely or unlikely.

8.1.3 Intuition: Coherence and Entropy

In a coherent argument, claims support each other, reducing uncertainty. In an incoherent argument, claims are scattered and contradictory, increasing uncertainty. We measure this via:

Total entropy: the combined uncertainty of all claims in the pitch.

A lower total entropy suggests a more coherent, confident argument. However, this must be balanced: we don't want an argument that is falsely confident. Calibration matters.

8.2 Mutual Information

8.2.1 Measuring Dependence

Definition 8.2 (Mutual Information)**.** *The mutual information between two random variables X and Y is:*

$$I(X;Y) = H(X) + H(Y) - H(X,Y)$$

where $H(X)$ is entropy of X, $H(Y)$ is entropy of Y, and $H(X,Y)$ is joint entropy. Alternatively:

$$I(X;Y) = \sum_{x,y} \Pr(x,y) \log \frac{\Pr(x,y)}{\Pr(x)\Pr(y)}$$

Mutual information ranges from 0 (independent variables) to $\min(H(X), H(Y))$ (fully dependent). It measures how much knowing X reduces uncertainty about Y.

8.2.2 Worked Example: Measuring Claim Dependencies

Let X be the claim "Market is growing rapidly" and Y be the claim "Revenue is growing rapidly." Assign probabilities:

	$Y = \top$	$Y = \bot$	$P(X)$
$X = \top$	0.4	0.1	0.5
$X = \bot$	0.2	0.3	0.5
$P(Y)$	0.6	0.4	

Compute entropies:

$$H(X) = -0.5\log(0.5) - 0.5\log(0.5) = 1 \text{ bit} \tag{37}$$
$$H(Y) = -0.6\log(0.6) - 0.4\log(0.4) \approx 0.971 \text{ bits} \tag{38}$$
$$H(X,Y) = -0.4\log(0.4) - 0.1\log(0.1) - 0.2\log(0.2) - 0.3\log(0.3) \tag{39}$$
$$\approx 0.971 - 0.330 - 0.464 + 0.521 = 1.698 \text{ bits} \tag{40}$$

Wait, let me recalculate $H(X,Y)$ correctly:

$$H(X,Y) = -\sum_{x,y} P(x,y)\log P(x,y) \tag{41}$$
$$= -[0.4\log(0.4) + 0.1\log(0.1) + 0.2\log(0.2) + 0.3\log(0.3)] \tag{42}$$
$$\approx -[-0.321 - 0.332 - 0.278 - 0.361] \tag{43}$$
$$\approx 1.292 \text{ bits} \tag{44}$$

Then mutual information:

$$I(X;Y) = 1 + 0.971 - 1.292 = 0.679 \text{ bits}$$

This positive mutual information indicates that knowing whether the market is growing significantly reduces uncertainty about whether revenue is growing. The claims are dependent, suggesting coherence.

8.3 Compression-Based Coherence

8.3.1 Intuition: Compressibility as a Coherence Signal

A coherent argument, when encoded in natural language, should be compressible: you can explain the entire narrative efficiently because claims build on each other.

Definition 8.3 (Compression Ratio)**.** *Given a text t (a pitch or argument), let:*

- *$L(t)$ be the compressed length of t using a standard compressor (e.g., LZMA, GZIP).*
- *$L(t_1) + L(t_2) + \cdots + L(t_n)$ be the sum of compressed lengths of individual sentences (or paragraphs).*

The compression ratio is:

$$R_c = \frac{L(t)}{L(t_1) + L(t_2) + \cdots + L(t_n)}$$

A ratio less than 1 indicates that the full text compresses better than the sum of parts, suggesting coherence (the claims form a unified whole).

8.3.2 Worked Example: Compression Analysis

Coherent pitch (all claims support a unified theme):

```
We target enterprise software (Claim 1).
Enterprise customers pay high prices (Claim 2).
Our product is 10x cheaper (Claim 3).
Therefore, we have strong competitive advantage (Claim 4).
```

Suppose:

- Compressed full text: 150 bytes
- Sum of compressed individual sentences: 120 + 95 + 105 + 110 = 430 bytes
- Compression ratio: $150/430 \approx 0.35$

Low ratio indicates high coherence: the full text compresses to 35% of the sum of parts because shared concepts (enterprise, pricing) are described once and referenced implicitly.

Incoherent pitch (claims scatter across topics):

```
We target enterprise software (Claim 1).
Artificial intelligence will transform medicine (Claim 2).
Cloud computing reduces infrastructure costs (Claim 3).
Our team includes Stanford PhDs (Claim 4).
```

Suppose:

- Compressed full text: 400 bytes (many independent topics)
- Sum of compressed individual sentences: 120 + 130 + 130 + 105 = 485 bytes
- Compression ratio: $400/485 \approx 0.82$

Higher ratio indicates lower coherence: the sentences don't share concepts or support a unified theme, so compression gains are minimal.

8.4 Redundancy

8.4.1 Measuring Repetition

Coherence involves a balance: claims should relate to each other but not be redundant.

Definition 8.4 (Redundancy)**.** *Redundancy is the extent to which information is repeated across claims. Formally:*

$$D_r = 1 - \frac{H(\textit{claims combined})}{H(\textit{average claim})}$$

High redundancy means claims largely repeat each other; low redundancy means they add new information.

8.4.2 The Coherence Sweet Spot

Optimal coherence involves a balance:

- **Low redundancy, high mutual information:** claims are diverse but mutually supporting. Example: "The market is large, we have product-market fit, our unit economics are positive." Each claim adds new information, but together they paint a coherent picture.
- **High redundancy:** claims mostly repeat. Example: "We are growing, we're expanding, we're scaling, the business is booming." Low coherence; the claims add little new value.
- **Low mutual information:** claims are independent. Example: "The market is large, our CEO has a PhD, the office is in Mountain View." No coherent narrative.

The coherence system targets the sweet spot: maximum mutual information with minimal redundancy.

9 Graph-Theoretic Measures

9.0.1 Arguments as Networks

We've modeled arguments as abstract argumentation frameworks (directed graphs with attack relations). Graph theory provides tools to measure structural properties that correlate with coherence.

9.1 Betweenness Centrality

9.1.1 Identifying Key Claims

Not all claims are equally important. Some claims support or attack many others, making them central to the argument's structure.

Definition 9.1 (Betweenness Centrality)**.** *For a node (claim) v in a directed graph, betweenness centrality is:*

$$C_B(v) = \sum_{s \neq v \neq t} \frac{\sigma(s,t|v)}{\sigma(s,t)}$$

where $\sigma(s,t)$ is the total number of shortest paths from s to t, and $\sigma(s,t|v)$ is the number of those paths that pass through v. Normalized betweenness divides by $(n-1)(n-2)/2$ for directed graphs.

Intuitively, a claim with high betweenness centrality lies on many shortest paths between other claims—it is crucial to the argument's structure.

9.1.2 Worked Example: Network Analysis of a Pitch

Consider a pitch with claims modeled as an argument graph:

- V_1: "Market is growing 30% annually."
- V_2: "Our product meets a key unmet need."
- V_3: "We have strong product-market fit."
- V_4: "Revenue is growing exponentially."
- V_5: "We will be profitable within 2 years."

Support edges ($\rightarrow$ indicates "supports"):

- $V_1 \rightarrow V_4$ (market growth supports revenue growth)
- $V_2 \rightarrow V_3$ (unmet need supports product-market fit)
- $V_3 \rightarrow V_4$ (product-market fit supports revenue growth)
- $V_4 \rightarrow V_5$ (revenue growth supports profitability)

The network:

```
V1 → \
        V4 → V5
V2 → V3 → /
```

Betweenness centrality:

- V_4 has high betweenness (lies on paths $V_1 \rightarrow V_5$ and $V_3 \rightarrow V_5$).
- V_5 has low betweenness (terminal node).
- V_3 has moderate betweenness (connects V_2 to V_4, V_5).

Interpretation: V_4 (exponential revenue growth) is central to the pitch's logic—it connects market opportunity to profitability. A vulnerability in this claim would undermine the entire argument.

9.2 Network Density

9.2.1 Measuring Overall Connectedness

Definition 9.2 (Network Density)**.** *For a directed graph with n nodes and m edges, density is:*

$$\rho = \frac{m}{n(n-1)}$$

ranging from 0 (no connections) to 1 (fully connected graph). Higher density indicates a more tightly integrated argument.

9.2.2 Example: Dense vs. Sparse Networks

A pitch with 5 claims and 8 support/attack edges:

$$\rho = \frac{8}{5 \times 4} = \frac{8}{20} = 0.4$$

This moderate density indicates that most claims relate to others, but the graph is not fully dense. A coherent pitch might have density 0.3–0.5 (claims are related but not redundant).

A pitch with 5 claims and only 2 edges:

$$\rho = \frac{2}{20} = 0.1$$

Low density suggests claims are isolated; the argument lacks narrative cohesion.

9.3 Circular Reasoning Detection

9.3.1 Identifying Logical Loops

Circular reasoning (begging the question) is a fallacy where a conclusion is used to support a premise that supposedly justifies the conclusion. In graph terms, this is a cycle in the support graph.

Definition 9.3 (Cycle Detection)**.** *A cycle in a directed graph is a path $v_1 \to v_2 \to \cdots \to v_k \to v_1$ where v_i are distinct.*

In an argumentation framework, a cycle of support edges indicates potential circular reasoning.

9.3.2 Example: Detecting Circularity

Suppose a pitch contains:

- A: "Our product is innovative because it uses cutting-edge AI."
- B: "Our AI is cutting-edge because our product is innovative."

This creates a cycle: $A \to B \to A$ (where $\to$ denotes "supports"). The coherence system detects this and flags it as circular reasoning. Both claims may be true, but they don't provide independent justification for each other.

The system differentiates:

- Vicious cycles (begging the question): low coherence.
- Benign cycles (mutual reinforcement among independent premises): acceptable.

A cycle is benign if the supporting claims have independent justification outside the cycle.

10 The Composite Coherence Score

10.0.1 Synthesizing Six Independent Measures

The coherence system measures coherence through six independent layers:

1. **Logical Consistency (Sections 2–3):** No contradictions in propositional/first-order logic or paraconsistent logic.
2. **Argument Structure (Section 4):** Dung-style abstract argumentation frameworks and ASPIC+ structured arguments.
3. **Natural Language Inference (Section 5):** NLI signals of entailment and contradiction between sentence pairs.
4. **Embedding Coherence (Section 6):** Semantic clustering, cosine similarity, and geometric contradiction detection.
5. **Probabilistic Coherence (Section 7):** Roche's measure of mutual support among claims.
6. **Information-Theoretic Coherence (Section 8):** Compression ratio, mutual information, and redundancy.
7. **Graph Structure (Section 9):** Network density, centrality, and circular reasoning.

None of these measures is perfect or sufficient on its own. The composite score is conservative: an argument must pass multiple independent tests to be deemed highly coherent. We use a weighted sum with careful attention to calibration.

10.0.2 Score Aggregation

Let the seven layer scores be $s_1, \ldots, s_7$, each normalized to $[0, 1]$ where 1 is most coherent. The composite score is:

$$\text{Coherence} = \sum_{i=1}^{7} w_i s_i$$

where weights w_i sum to 1. We recommend:

- $w_1 = 0.25$ (logical consistency is foundational)
- $w_2 = 0.20$ (argument structure)
- $w_3 = 0.15$ (NLI)
- $w_4 = 0.15$ (embedding coherence)
- $w_5 = 0.15$ (probabilistic)
- $w_6 = 0.10$ (information-theoretic)

These weights reflect the relative importance and confidence in each measure. Logical consistency is weighted highest because it is foundational—a logically inconsistent argument is incoherent, regardless of other signals.

10.0.3 Threshold and Interpretation

Coherence scores typically range from 0 to 1 (though some measures can exceed 1; we clamp to 1). Interpretation:

- Coherence > 0.75: High coherence. Argument is well-structured, logically sound, and narratively unified. Suitable for further analysis.
- $0.5 <$ Coherence ≤ 0.75: Moderate coherence. Argument has merit but may contain tensions, logical gaps, or scattered themes. Requires clarification.
- $0.25 <$ Coherence ≤ 0.5: Low coherence. Argument is fragmented, contains apparent contradictions, or jumps between unrelated claims. Signals either poor reasoning or lack of preparation.
- Coherence ≤ 0.25: Very low coherence. Argument is incoherent; claims contradict or scatter across unrelated topics. Suggests the proposer has not thought through the value thesis.

10.0.4 Confidence Intervals

Because each layer score has uncertainty (due to model limitations, ambiguity in natural language, etc.), we report confidence intervals:

$$\text{Coherence} = \bar{s} \pm \sigma$$

where $\bar{s}$ is the mean score and σ is the standard deviation across the seven layers. If σ is large, the argument scores inconsistently across layers, suggesting that different aspects measure different coherence dimensions (e.g., logically sound but narratively scattered).

11 Algorithms

We now present the core algorithms for computing coherence scores at each layer.

11.0.1 Algorithm 1: Logical Consistency Checking

Input: `Set of propositional formulas` $\Phi = \{\phi_1, \ldots, \phi_n\}$
Output: `Boolean: true if` Φ `is satisfiable, false if unsatisfiable`

```
Input preprocessing: Convert each natural language claim to a propositional formula.
SAT solver: Use a modern SAT solver (e.g., Z3, CaDiCaL) to determine if Φ is
satisfiable.
return true if satisfiable, false if unsatisfiable.   ▷ Time complexity: NP-complete, but
practical solvers are fast for typical argument sizes.
```

Algorithm 1: Check Propositional Consistency

11.0.2 Algorithm 2: Computing Dung Extensions

Input: `Argumentation framework` (A, R) `where` A `is arguments,` R `is attack relation`
Output: `Set of acceptable arguments (grounded extension)`

```
Initialize S := ∅ (the acceptable set)
Initialize U := A (unattacked arguments)
Repeat until fixed point:
  Find all arguments in U that are unattacked by U.
  Add these to S.
  Remove attacked arguments from U.
return S as the grounded extension.
```
▷ *Time complexity:* $O(n^2)$ *where* $n = |A|$. *Polynomial for all semantics.*

Algorithm 2: Compute Grounded Extension (Dung Semantics)

11.0.3 Algorithm 3: NLI Scoring

Input: `Set of sentences` $S = \{s_1, \ldots, s_m\}$`; NLI model` M
Output: `Matrix of entailment/neutral/contradiction relations`

```
Initialize result matrix R of size m × m.
For each pair (s_i, s_j) with i ≠ j:
  Invoke M(s_i, s_j) to get probabilities Pr(ent), Pr(neut), Pr(cont).
  Classify as entailment/neutral/contradiction based on highest probability.
  Store probabilities in R[i][j].
return R.
```
▷ *Time complexity:* $O(m^2 \cdot T)$ *where* T *is inference time of NLI model.*

Algorithm 3: Compute Pairwise NLI Relations

11.0.4 Algorithm 4: Embedding Coherence

Input: `Set of sentences` $S = \{s_1, \ldots, s_m\}$`; embedding model` emb
Output: `Coherence score reflecting semantic clustering`

```
Compute embeddings e_i := emb(s_i) for all sentences.
Compute pairwise cosine similarities C[i][j] := cos(e_i, e_j).
Compute average pairwise similarity: μ_C := 1/(m(m−1)) Σ_{i≠j} C[i][j].
Compute variance: σ_C^2 := 1/(m(m−1)) Σ_{i≠j} (C[i][j] − μ_C)^2.
Coherence score: score := μ_C (higher average similarity = more coherence).
return score, along with variance as confidence measure.
```
▷ *Time complexity:* $O(m^2 \cdot d)$ *where* d *is embedding dimension.*

Algorithm 4: Compute Embedding-Based Coherence

11.0.5 Algorithm 5: Probabilistic Coherence (Roche's Measure)

Input: `Marginal probabilities` $\Pr(C_i)$ `and joint probability` $\Pr(\bigwedge_i C_i)$
Output: `Roche coherence score`

`Compute sum of marginals:` $\Sigma := \sum_i \Pr(C_i)$.
`Compute joint:` $J := \Pr(\bigwedge_i C_i)$.
`Roche score:` $R := \frac{n \cdot J - \Sigma}{\Sigma}$ `where` $n =$ `number of claims.`
`return` R. ▷ *Score interpretation:* $R > 0$ *indicates positive coherence.*

Algorithm 5: Compute Roche Coherence

11.0.6 Algorithm 6: Compression Ratio

Input: `Full argument text` t`; individual claim texts` $t_1, \ldots, t_n$
Output: `Compression ratio coherence score`

`Compress full text:` $L_{\texttt{full}} := \texttt{compress}(t)$.
`Compress individual claims:` $L_i := \texttt{compress}(t_i)$ `for each` i.
`Compute sum:` $L_{\texttt{sum}} := \sum_i L_i$.
`Compression ratio:` $R_c := L_{\texttt{full}} / L_{\texttt{sum}}$.
`Coherence score:` $\texttt{score} := 1 - R_c$ `(values less than 1 map to positive coherence).`
`return score.` ▷ *Intuitively: low* R_c *(much compression) indicates high coherence.*

Algorithm 6: Compute Compression-Based Coherence

12 Conceptual Ontology and Philosophical Cohesion

The six-layer coherence score measures whether a body of text is *internally consistent*. But philosophical cohesion requires something deeper: the argument must not merely avoid contradictions—it must form a *unified conceptual structure* in which claims are systematically related through shared concepts, and those concepts occupy defensible positions within a broader ontology of ideas.

This section formalizes the notion of a **conceptual ontology** and develops mathematical tools for measuring how well an argument's claims cohere at the level of the concepts they invoke.

12.1 Conceptual Ontologies

Definition 12.1 (Conceptual Ontology)**.** *A* ***conceptual ontology*** *is a directed acyclic graph* $\mathcal{O} = (V, E, \tau)$ *where:*

- V *is a set of* ***concepts*** *(e.g., "market," "value," "efficiency," "rationality").*
- $E \subseteq V \times V$ *is a set of directed edges representing* ***subsumption*** *or* ***dependency*** *relations:* $(v_i, v_j) \in E$ *means concept* v_i *is a specialization of or depends on concept* v_j*.*
- $\tau : V \to \{0, 1, 2, \ldots\}$ *assigns a* ***depth level*** *to each concept, where* $\tau(v) = 0$ *for foundational concepts (the roots of the DAG) and* $\tau(v) > 0$ *for derived concepts.*

Example. *A partial ontology for the argument "Value is persuasion" might include:*

$$V = \{\textit{Logic}, \textit{Persuasion}, \textit{Value}, \textit{Markets}, \textit{Coherence}, \textit{Truth}, \textit{Efficiency}, \textit{Allocation}\}$$
$$E \ni (\textit{Coherence}, \textit{Logic}),\ (\textit{Value}, \textit{Persuasion}),\ (\textit{Efficiency}, \textit{Allocation}),\ (\textit{Markets}, \textit{Allocation}),$$
$$(\textit{Persuasion}, \textit{Logic}),\ (\textit{Truth}, \textit{Coherence})$$

Here Coherence depends on Logic, Value depends on Persuasion, and so forth. The depth levels: $\tau(\textit{Logic}) = 0$, $\tau(\textit{Persuasion}) = 1$, $\tau(\textit{Value}) = 2$.

12.2 Concept Grounding

Every claim in a text invokes certain concepts from the ontology. We formalize this as a **grounding function**.

Definition 12.2 (Concept Grounding)**.** *Given a set of claims* $\{C_1, \ldots, C_n\}$ *and an ontology* $\mathcal{O} = (V, E, \tau)$*, the* ***grounding function*** *is:*

$$g : \{C_1, \ldots, C_n\} \to \mathcal{P}(V)$$

mapping each claim to the set of concepts it invokes. In practice, g *is computed by:*

1. *Named entity and noun phrase extraction from each claim.*
2. *Mapping extracted terms to ontology nodes via embedding similarity:*

$$v^* = \arg\max_{v \in V} \ \textit{cosim}(\textit{embed}(\textit{term}), \textit{embed}(v))$$

with a threshold θ *(typically* $\theta = 0.65$*).*

12.3 Ontological Coverage

A coherent philosophical argument should have broad, connected coverage of its relevant ontology—not isolated clusters of concepts with gaps between them.

Definition 12.3 (Ontological Coverage)**.** *The* ***coverage*** *of an argument with respect to an ontology is:*

$$\kappa(\{C_i\}, \mathcal{O}) = \frac{|\bigcup_{i=1}^{n} g(C_i)|}{|V|}$$

This is the fraction of ontology concepts that the argument addresses.

But raw coverage is insufficient. An argument that mentions every concept superficially is less coherent than one that deeply develops a connected subset. We therefore define **connected coverage**.

Definition 12.4 (Connected Coverage)**.** *Let* $G = (V_g, E_g)$ *be the subgraph of* $\mathcal{O}$ *induced by the grounded concepts* $V_g = \bigcup_i g(C_i)$*. The* ***connected coverage*** *is:*

$$\kappa_c(\{C_i\}, \mathcal{O}) = \frac{|V_{\max}|}{|V_g|}$$

where $V_{\max}$ *is the largest connected component of* G *(treating edges as undirected for this purpose). A value of* 1.0 *means all invoked concepts form a single connected cluster; lower values indicate conceptual fragmentation.*

12.4 Depth-Weighted Coherence

Not all concepts are equal. Foundational concepts (at $\tau = 0$) carry more weight than peripheral ones. An argument that is coherent at the foundational level but has surface-level contradictions is philosophically stronger than one that is superficially consistent but contradicts itself at the foundations.

Definition 12.5 (Depth-Weighted Layer Score)**.** *For any layer score s_i, we can compute a depth-weighted variant by restricting the computation to claims that invoke concepts at a given depth. Let $C^{(\ell)} = \{C_j : \exists v \in g(C_j), \tau(v) = \ell\}$ be the claims that invoke concepts at depth ℓ. Then:*

$$s_i^{dw} = \frac{\sum_{\ell=0}^{L} \gamma^\ell \cdot s_i(C^{(\ell)})}{\sum_{\ell=0}^{L} \gamma^\ell}$$

*where $\gamma \in (0, 1)$ is a **depth discount factor** (typically $\gamma = 0.7$) that assigns exponentially more weight to foundational levels. Here $s_i(C^{(\ell)})$ denotes the layer-i score computed on only the claims at depth ℓ.*

When $\gamma = 0.7$ and $L = 3$:

- Depth 0 (foundations): weight 1.000 (normalized: 0.395)
- Depth 1: weight 0.700 (normalized: 0.276)
- Depth 2: weight 0.490 (normalized: 0.194)
- Depth 3: weight 0.343 (normalized: 0.135)

This ensures that a contradiction at the foundational level (e.g., simultaneously asserting that value is objective and that value is purely subjective) penalizes the score far more than a surface-level inconsistency.

12.5 Conceptual Dependency Satisfaction

If concept A depends on concept B in the ontology, then any claim about A presupposes that B has been adequately addressed. An argument that discusses "market efficiency" without grounding "rationality" or "information" has an unsatisfied dependency.

Definition 12.6 (Dependency Satisfaction)**.** *For each grounded concept $v \in V_g$, let $deps(v) = \{u : (v, u) \in E\}$ be its parents in the ontology. The **dependency satisfaction ratio** is:*

$$\delta(\{C_i\}, \mathcal{O}) = \frac{\sum_{v \in V_g} |deps(v) \cap V_g|}{\sum_{v \in V_g} |deps(v)|}$$

A value of 1.0 means every dependency of every invoked concept is also addressed by the argument. Lower values indicate gaps—the argument is building on concepts it has not adequately justified.

12.6 The Philosophical Cohesion Score

We now combine these ontological measures into a single **philosophical cohesion** score that augments the original composite coherence score.

Definition 12.7 (Philosophical Cohesion)**.**

$$\mathcal{P}(\{C_i\}, \mathcal{O}) = \lambda_1 \cdot \kappa_c + \lambda_2 \cdot \delta + \lambda_3 \cdot \bar{s}^{dw}$$

where:

- κ_c *is the connected coverage score.*
- δ *is the dependency satisfaction ratio.*
- $\bar{s}^{dw} = \sum_i w_i \, s_i^{dw}$ *is the depth-weighted composite coherence score.*
- $\lambda_1 = 0.20$, $\lambda_2 = 0.30$, $\lambda_3 = 0.50$ *are the cohesion weights.*

The dependency satisfaction weight is highest among the ontological components ($\lambda_2 = 0.30$) because an argument with ungrounded assumptions is the most common failure mode in philosophical reasoning.

12.7 The Extended Composite Score

The original coherence score and the philosophical cohesion score can be combined:

$$\mathcal{C}_{\text{ext}}(T) = \beta \cdot \mathcal{C}(T) + (1 - \beta) \cdot \mathcal{P}(\{C_i\}, \mathcal{O})$$

where $\beta \in [0, 1]$ controls the balance between raw coherence and philosophical depth. For startup pitch analysis, $\beta \approx 0.7$ (raw coherence matters most). For philosophical argument evaluation, $\beta \approx 0.4$ (conceptual structure matters more).

13 Cross-Domain Coherence Transfer

A powerful philosophical argument draws connections across domains—linking economics to ethics, logic to psychology, biology to politics. The coherence engine must measure not only within-domain consistency but also **cross-domain bridging**: does the argument successfully connect ideas from different fields into a unified framework?

13.1 Domain Partitions

Definition 13.1 (Domain Partition)**.** *A **domain partition** of a conceptual ontology $\mathcal{O} = (V, E, \tau)$ is a partition $V = D_1 \sqcup D_2 \sqcup \cdots \sqcup D_m$ where each D_k corresponds to a distinct intellectual domain (e.g., economics, ethics, logic, psychology).*

Definition 13.2 (Cross-Domain Bridge)**.** *A claim C_j is a **cross-domain bridge** if $g(C_j) \cap D_k \neq \varnothing$ and $g(C_j) \cap D_\ell \neq \varnothing$ for some $k \neq \ell$. That is, the claim invokes concepts from at least two different domains.*

Definition 13.3 (Bridge Density)**.**

$$\beta_d(\{C_i\}) = \frac{|\{C_j : C_j \text{ is a cross-domain bridge}\}|}{n}$$

This is the fraction of claims that serve as bridges between domains.

13.2 Inter-Domain Coherence Matrix

For each pair of domains (D_k, D_ℓ), we compute the average embedding coherence between claims grounded in those domains:

$$M_{k\ell} = \frac{1}{|B_{k\ell}|} \sum_{(i,j)\in B_{k\ell}} \text{cosim}(\mathbf{e}_i, \mathbf{e}_j)$$

where $B_{k\ell} = \{(i,j) : g(C_i) \cap D_k \neq \varnothing,\ g(C_j) \cap D_\ell \neq \varnothing,\ k \neq \ell\}$.

The matrix M is symmetric. Its minimum eigenvalue $\lambda_{\min}(M)$ indicates the weakest link between any pair of domains. A positive minimum eigenvalue means all domain-pairs are at least somewhat coherent; a near-zero eigenvalue indicates a domain that is disconnected from the rest of the argument.

13.3 Cross-Domain Coherence Score

Definition 13.4 (Cross-Domain Score)**.**

$$s_{cross} = \frac{\beta_d \cdot \bar{M}}{1 + \sigma_M}$$

where $\bar{M}$ is the mean of the off-diagonal elements of M and σ_M is their standard deviation. This rewards high bridge density and high average inter-domain coherence, while penalizing uneven bridging (high variance means some domain pairs are well-connected while others are isolated).

14 Abductive Coherence: Explanatory Power

The measures above assess whether claims are *consistent* and *connected*. But philosophical arguments do more than avoid contradictions—they *explain*. An argument is stronger when its claims collectively provide a good explanation for observed phenomena.

14.1 The Explanatory Coverage Problem

Definition 14.1 (Explananda and Explanantia)**.** *Given a text T, we distinguish:*

- ***Explananda*** $\mathcal{E} = \{E_1, \ldots, E_p\}$*: phenomena the argument seeks to explain (e.g., "why do some businesses fail?").*
- ***Explanantia*** $\mathcal{X} = \{X_1, \ldots, X_q\}$*: explanatory claims the argument offers (e.g., "businesses fail because their value propositions contain internal contradictions").*

An ***explanation relation*** $\models_{exp}$ *holds when a subset of explanantia logically or probabilistically accounts for an explanandum:*

$$\{X_{i_1}, \ldots, X_{i_k}\} \models_{exp} E_j$$

14.2 Explanatory Coverage

Definition 14.2 (Explanatory Coverage)**.**

$$\epsilon = \frac{|\{E_j \in \mathcal{E} : \exists \text{ subset of } \mathcal{X} \text{ that explains } E_j\}|}{|\mathcal{E}|}$$

This is the fraction of stated explananda that the argument's explanantia can account for.

14.3 Explanatory Parsimony

An argument that requires 50 claims to explain 3 phenomena is less parsimonious than one that explains the same phenomena with 5 claims. Following Occam's Razor:

Definition 14.3 (Parsimony Score)**.**

$$\pi = 1 - \frac{\bar{k}}{q}$$

where $\bar{k}$ is the average number of explanantia needed per explanandum and q is the total number of explanantia. A score near 1 means each explanation is economical; near 0 means the argument needs almost all of its claims to explain anything.

14.4 The Abductive Coherence Score

Definition 14.4 (Abductive Coherence)**.**

$$s_{abd} = \epsilon \cdot \pi$$

This is the product of coverage and parsimony. An argument that explains everything it sets out to explain, using few claims per explanation, achieves a high abductive coherence score.

15 Temporal Coherence Dynamics

Arguments evolve. A philosophical framework refined over time should become *more* coherent, not less. This section formalizes the measurement of coherence trajectories.

15.1 The Coherence Trajectory

Definition 15.1 (Coherence Trajectory)**.** *Given a sequence of argument versions $T^{(1)}, T^{(2)}, \ldots, T^{(K)}$ observed at times $t_1 < t_2 < \cdots < t_K$, the **coherence trajectory** is the function:*

$$\gamma : \{t_1, \ldots, t_K\} \to [0, 1], \quad \gamma(t_k) = \mathcal{C}(T^{(k)})$$

Definition 15.2 (Coherence Velocity)**.** *The **coherence velocity** at time t_k is the discrete derivative:*

$$v(t_k) = \frac{\gamma(t_k) - \gamma(t_{k-1})}{t_k - t_{k-1}}$$

Positive velocity indicates improvement; negative velocity indicates degradation.

Definition 15.3 (Coherence Momentum)**.** *The **coherence momentum** is the exponentially weighted moving average of velocity:*

$$\mu(t_k) = \alpha \cdot v(t_k) + (1 - \alpha) \cdot \mu(t_{k-1})$$

with smoothing parameter $\alpha \in (0, 1)$ (typically $\alpha = 0.3$). Positive momentum predicts continued improvement; negative momentum predicts continued degradation.

15.2 Convergence and Divergence

Definition 15.4 (Coherence Convergence)**.** *An argument sequence* ***converges*** *if $\gamma(t_k)$ is monotonically nondecreasing for all $k \geq k_0$ for some k_0. The* ***convergence rate*** *is:*

$$r = -\log\left(\frac{1-\gamma(t_K)}{1-\gamma(t_{k_0})}\right) \cdot \frac{1}{K-k_0}$$

Higher r means faster convergence toward perfect coherence.

This provides a formal basis for evaluating whether an intellectual project (a book, a business thesis, a philosophical framework) is progressing toward internal unity or fragmenting over time.

16 Formal Properties and Theorems

We establish several formal properties of the extended coherence framework.

Theorem 16.1 (Monotonicity of Contradiction Removal)**.** *If $\Phi' = \Phi \setminus \{\phi\}$ where ϕ is a member of some MUS of Φ, then $s_1(\Phi') \geq s_1(\Phi)$. That is, removing a contradictory claim never decreases the logical consistency score.*

Proof. $|\mathrm{MUS}(\Phi')| \leq |\mathrm{MUS}(\Phi)|$ (removing a MUS member dissolves at least that MUS), and $|\Phi'| = |\Phi| - 1$. Therefore:

$$s_1(\Phi') = 1 - \frac{|\mathrm{MUS}(\Phi')|}{|\Phi'|} \geq 1 - \frac{|\mathrm{MUS}(\Phi)|}{|\Phi|-1} \geq 1 - \frac{|\mathrm{MUS}(\Phi)|}{|\Phi|} = s_1(\Phi)$$

The second inequality holds when $|\mathrm{MUS}(\Phi)| < |\Phi|$, which is true whenever Φ contains any non-contradictory claims. □

Theorem 16.2 (Grounded Extension Existence and Uniqueness)**.** *For any argumentation framework $AF = (\mathcal{A}, \mathcal{R})$, the grounded extension E_g exists and is unique.*

Proof. By Dung's fundamental lemma, the characteristic function F is monotone with respect to set inclusion. By the Knaster–Tarski fixed-point theorem, every monotone function on a complete lattice has a least fixed point. The powerset $\mathcal{P}(\mathcal{A})$ ordered by inclusion is a complete lattice. Therefore F has a unique least fixed point, which is the grounded extension E_g. □

Theorem 16.3 (Coherence Under Refinement)**.** *Let $T^{(k+1)}$ be obtained from $T^{(k)}$ by either (a) resolving a contradiction (removing or modifying a claim in some MUS), or (b) adding a claim that is entailed by existing claims. Then $\mathcal{C}(T^{(k+1)}) \geq \mathcal{C}(T^{(k)})$ provided the modification does not introduce new contradictions.*

Proof. (a) Contradiction resolution: by the monotonicity theorem, s_1 does not decrease. The NLI layer s_3 also cannot decrease (average contradiction probability is reduced). Layers s_2, s_4, s_5, s_6 are unaffected or improved. Hence $\mathcal{C}$ is nondecreasing.

(b) Adding an entailed claim: the new claim is consistent with existing claims (s_1 unchanged), supported by the argumentation structure (s_2 unchanged or increased), has low NLI contradiction probability with existing claims (s_3 unchanged or increased), is semantically similar to the claims that entail it (s_4 increased), and increases the joint probability (s_5 increased). Compression improves due to shared content (s_6 increased). Hence $\mathcal{C}$ is nondecreasing. □

Theorem 16.4 (Sparsity Separation)**.** *Let $\mathbf{d}_c$ be a contradiction difference vector and $\mathbf{d}_e$ be an entailment difference vector produced by the same embedding model. Then with probability at least $1-\epsilon$ over the choice of sentence pairs from a natural language distribution:*

$$H(\mathbf{d}_c) > H(\mathbf{d}_e) + \Delta$$

for some model-dependent margin $\Delta > 0$. That is, contradiction difference vectors are strictly sparser than entailment difference vectors.

Proof sketch. Entailment preserves topic *and* adds information along the same semantic subspace, yielding a difference vector whose energy is distributed across many dimensions of the shared subspace. Contradiction preserves topic but *flips* a small number of semantic features (negation, antonymy), concentrating the difference in those specific dimensions. By the SparseCL characterization (Xu et al., 2024), the flipped features constitute approximately 14% of the embedding dimensions for standard sentence-transformer models, yielding Hoyer indices of ~ 0.44 for contradictions vs. ~ 0.21 for entailments—a margin of $\Delta \approx 0.23$. □

17 Algorithm: Full Extended Coherence Pipeline

We present the complete algorithm for computing the extended coherence score with philosophical cohesion.

Algorithm 7: Extended Coherence Pipeline
Input: Text T, conceptual ontology $\mathcal{O} = (V, E, \tau)$, domain partition $\{D_k\}$, parameters β, γ, λ_i.
Output: Extended coherence score $\mathcal{C}_{\text{ext}}(T)$.

1. **Parse**: Extract claims $C_1, \ldots, C_n$ from T using NLP (sentence segmentation, coreference resolution, clause extraction).
2. **Ground**: For each C_i, compute grounding $g(C_i) \subseteq V$ via embedding similarity to ontology concepts.
3. **Compute six-layer coherence**: Execute Layers 1–6 as specified in Section 10 to obtain $s_1, \ldots, s_6$ and composite $\mathcal{C}(T)$.
4. **Compute depth-weighted variants**: For each layer i and each depth ℓ, compute $s_i(C^{(\ell)})$ on the claims at that depth. Aggregate using γ discounting.
5. **Ontological metrics**:
   (a) Connected coverage κ_c: build induced subgraph, find largest component.
   (b) Dependency satisfaction δ: check which concept dependencies are met.
6. **Philosophical cohesion**: $\mathcal{P} = \lambda_1 \kappa_c + \lambda_2 \delta + \lambda_3 \bar{s}^{\text{dw}}$.
7. **Cross-domain bridging**: Compute bridge density β_d, inter-domain matrix M, cross-domain score s_{cross}.
8. **Abductive coherence**: Identify explananda and explanantia, compute ϵ, π, s_{abd}.
9. **Extended score**:

$$\mathcal{C}_{\text{ext}}(T) = \beta \cdot \mathcal{C}(T) + (1-\beta) \cdot \left[\frac{\mathcal{P} + s_{\text{cross}} + s_{\text{abd}}}{3}\right]$$

Complexity: $O(n^2 d + n|V| + |V|^2)$ where n = number of claims, d = embedding dimension, $|V|$ = ontology size. The quadratic terms dominate for large arguments; in practice, $n < 500$ and $|V| < 1000$, making the full pipeline tractable.

18 Conjectures and Open Problems

This framework is a work in progress. Several open questions remain:

Conjecture 18.1 (Equivalence of Coherence Measures)**.** *Do all seven layer measures asymptotically converge on the same ranking of arguments? That is, if argument A is more coherent than argument B according to layer 1, do all layers agree?*

This is unknown. Different coherence notions (logical consistency vs. semantic clustering) may legitimately disagree.

Conjecture 18.2 (Calibration of Weights)**.** *The weights w_i for layer aggregation are currently set heuristically. Can they be learned from data? Specifically, given a dataset of arguments with human coherence judgments, can we optimize the weights to maximize agreement with human assessment?*

This would require a corpus of startup pitches or similar arguments with coherence ratings, which is not yet available.

Conjecture 18.3 (Domain Adaptation)**.** *Different argument domains (startup pitches, scientific papers, political speeches) may have different coherence signatures. Does a model trained on startup pitches generalize to scientific papers?*

Preliminary evidence suggests domain effects are significant. A domain-specific model would likely outperform a general model, but domain adaptation remains an open challenge.

Conjecture 18.4 (Temporal Coherence)**.** *This framework measures coherence at a snapshot in time. But arguments evolve: a founder refines the pitch, new market data arrives, contradictions are resolved. How should coherence be measured over time? Is an argument that evolves toward greater coherence "better" than one that is coherent from the start?*

This suggests a dynamic model of coherence, which we have not yet developed.

Conjecture 18.5 (Coherence and Truth)**.** *Is coherence correlated with truth? That is, do more coherent arguments tend to be true, and incoherent arguments tend to be false?*

This is the deepest question. Coherence is not sufficient for truth (a coherent fiction is still false), but it may be necessary (incoherent claims often signal confused thinking or deception). Empirical validation is needed.

Remark (A Word on Limitations)**.** *This framework is mathematical and formal, but natural language remains irreducibly ambiguous. No formal system can perfectly capture the nuances of human reasoning. The coherence engine should be seen as a tool for structuring analysis and catching obvious contradictions, not as an oracle of truth.*

Moreover, the framework assumes that coherence is desirable. But sometimes incoherence signals intellectual honesty: acknowledging that a problem is complex, that contradictions exist, that we don't fully understand something. An over-confident, false-coherent argument may be worse than a confused but genuine one.

The goal of the framework is to measure a specific property (internal logical consistency and narrative unity) that is useful for assessing argument quality, while remaining humble about what that measurement can and cannot tell us.